

Hariton Pitsik

ML ENGINEER · AI RESEARCHER · DATA SCIENTIST

Saratov, Russia

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Education

Saratov State University

BACHELOR | FUNDAMENTAL COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Saratov, Russia

September 2023 - June 2027

Skills

Programming languages

Python, Rust, SQL

Spoken languages

Russian (native), English (B1-B2)

Technologies

Git & GitHub, Docker, Linux, LaTeX, Typst, Jupyter, NumPy, Pandas, Matplotlib & Seaborn, Scikit-learn, XGBoost, CatBoost, FastAPI, ClearML, Streamlit, PyTorch, Transformers, PostgreSQL

Concepts

Linear Algebra, Mathematical Analysis, Probability Theory & Statistics, Computer Science, Classical ML Algorithms (Supervised & Unsupervised), Deep Learning, Natural Language Processing (NLP), Large Language Models (LLM)

Some achievements

Teaching & Social

- Participant of Development Students Club (DSC)
- I believe education is important and should be shared. I teach machine learning as part of a DSC community.
 - Lecture recordings on NumPy ([YouTube](#)) and Pandas ([YouTube](#)).
 - Machine Learning Club playlist: ([YouTube](#))

Competitions

- Participant in the HSE & X5 Growth Gradient contest, focused on time series forecasting ([Link](#)).
- Participant in the FSP hackathon as an analyst of existing hashing methods and potential sources of entropy.

Projects

Local RAG System

🌐 github.com/haritonn/pdf_rag

- Built a local RAG system with a switchable LLM backend;
- Implemented hybrid retrieval, document chunking, and automated relevance evaluation;
- Integrated the Qdrant vector database and built a Streamlit interface.

Transformers from Scratch

🌐 github.com/haritonn/transformers-scratch

- Implemented the full encoder-decoder architecture in PyTorch;
- Built a modular training pipeline for seq2seq tasks;
- Designed a fully customizable pipeline for experiments.

Caption Generator

🌐 github.com/haritonn/caption_gen

- Built an image captioning pipeline with a ResNet backbone and an attention-based LSTM;
- Implemented the project entirely in PyTorch without pre-built components;
- Implemented beam search and greedy decoding;

- Built an end-to-end training workflow.

ResNet with Attention

 github.com/haritonn/resnet_attention

- Implemented baseline ResNet and attention-augmented ResNet variants in PyTorch;
- Compared convergence speed and classification accuracy;
- Tracked experiments with ClearML.

Coursework

2nd Year: Generative Computer Vision

 github.com/haritonn/coursework2

COMPARED DIFFUSION MODELS, GANS, AND LARGE MULTIMODAL MODEL (LMM)-BASED APPROACHES FOR VIRTUAL TRY-ON. EVALUATED VISUAL QUALITY AND INFERENCE TIME ON A CUSTOM DATASET.

3rd Year: Analysis of Geometric Properties of Embeddings

ONGOING COURSEWORK FOCUSED ON THE ANALYSIS OF EMBEDDING SPACES.